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Java SE 27 & JDK 27

Class VolatileCallSite

java.lang.Object
    java.lang.invoke.CallSite
        java.lang.invoke.VolatileCallSite

```
public non-sealed class VolatileCallSite
extends CallSite
```

A `VolatileCallSite` is a `CallSite` whose target acts like a volatile variable. An `invokedynamic` instruction linked to a `VolatileCallSite` sees updates to its call site target immediately, even if the update occurs in another thread. There may be a performance penalty for such tight coupling between threads.

Unlike `MutableCallSite`, there is no syncAll operation on volatile call sites, since every write to a volatile variable is implicitly synchronized with reader threads.

In other respects, a `VolatileCallSite` is interchangeable with `MutableCallSite`.

Since:

1.7

See Also:

`MutableCallSite`

Constructor Summary

Constructors

Constructor	Description
VolatileCallSite(MethodHandle target)	Creates a call site with a volatile binding to its target.
VolatileCallSite(MethodType type)	Creates a call site with a volatile binding to its target.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
final MethodHandle	**dynamicInvoker**()	Produces a method handle equivalent to an invokedynamic instruction which has

		been linked to this call site.
final MethodHandle	**getTarget**()	Returns the target method of the call site, which behaves like a `volatile` field of the `VolatileCallSite`.
void	**setTarget**(MethodHandle newTarget)	Updates the target method of this call site, as a volatile variable.

Methods declared in class CallSite

type

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

VolatileCallSite

public VolatileCallSite(MethodType type)

Creates a call site with a volatile binding to its target. The initial target is set to a method handle of the given type which will throw an `IllegalStateException` if called.

Parameters:

type - the method type that this call site will have

Throws:

NullPointerException - if the proposed type is null

VolatileCallSite

public VolatileCallSite(MethodHandle target)

Creates a call site with a volatile binding to its target. The target is set to the given value.

Parameters:

`target` - the method handle that will be the initial target of the call site

Throws:

`NullPointerException` - if the proposed target is null

Method Details

getTarget

```
public final MethodHandle getTarget()
```

Returns the target method of the call site, which behaves like a `volatile` field of the `VolatileCallSite`.

The interactions of `getTarget` with memory are the same as of a read from a `volatile` field.

In particular, the current thread is required to issue a fresh read of the target from memory, and must not fail to see a recent update to the target by another thread.

Specified by:

`getTarget` in class `CallSite`

Returns:

the linkage state of this call site, a method handle which can change over time

See Also:

`setTarget(MethodHandle)`

setTarget

```
public void setTarget(MethodHandle newTarget)
```

Updates the target method of this call site, as a volatile variable. The type of the new target must agree with the type of the old target.

The interactions with memory are the same as of a write to a volatile field. In particular, any threads is guaranteed to see the updated target the next time it calls `getTarget`.

Specified by:

`setTarget` in class `CallSite`

Parameters:

newTarget - the new target

Throws:

NullPointerException - if the proposed new target is null

WrongMethodTypeException - if the proposed new target has a method type that differs from the previous target

See Also:

getTarget()

dynamicInvoker

```
public final MethodHandle dynamicInvoker()
```

Produces a method handle equivalent to an invokedynamic instruction which has been linked to this call site.

This method is equivalent to the following code:

```
MethodHandle getTarget, invoker, result;
getTarget = MethodHandles.publicLookup().bind(this, "getTarget", MethodType.
invoker = MethodHandles.exactInvoker(this.type());
result = MethodHandles.foldArguments(invoker, getTarget)
```

Specified by:

dynamicInvoker in class CallSite

Returns:

a method handle which always invokes this call site's current target

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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